



Construction of locally conservative fluxes for high order continuous Galerkin finite element methods

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ABSTRACT

Despite their robustness, it is known that standard continuous Galerkin Finite Element Methods (CGFEMs) do not produce a locally conservative flux field. As a result, their application to solving model problems that are derived from conservation laws can be limited. To remedy this issue some form of post-processing must be performed on the CGFEM solution. In this work, a simple post-processing technique is proposed to obtain a locally conservative flux field from a CGFEM solution. One distinct advantage of the proposed method is that it produces continuous normal flux at the element's boundary. The post-processing is implemented on nodal-centered control volumes that are constructed from the original finite element mesh. The post-processing method is performed by solving an independent set of low dimensional problems posed on each element. The associated linear algebra systems are of dimension $\frac{1}{2}(k+1)(k+2)$ where k is the polynomial degree of CGFEM basis on a triangular mesh. A theoretical investigation is conducted to confirm that the post-processed solution converges in an optimal fashion to the true solution in the H^1 semi-norm. Various numerical examples that demonstrate the performance of technique are given. Specifically, a simulation of a model for single-phase flow in a heterogeneous system is presented to show the necessity of the local conservation as well as the effective performance of the post-processing technique.

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1. Introduction

A wide variety of initial/boundary value problems can be solved using the finite volume element method (FVEM) or continuous Galerkin finite element method (CGFEM). In both methods approximate solutions are represented as linear combinations of the finite element basis functions. This representation of the approximate solution gives these methods the ability to solve problems that are posed on complicated geometries. However, the variational formulations used to find approximate solutions with each method are remarkably different. The CGFEM is defined using a global variational formulation while the FVEM relies on a local variational formulation, namely one that imposes local conservation of the approximate solution's flux. Unfortunately, the nature of the FVEM formulation does not lend itself well to rigorous analysis and error estimation. In the case of linear finite element, it is well-known that the bilinear form of FVEM is closely related to its CGFEM counterpart, and this closeness is exploited to carry out the error analysis of FVEM [1,2]. Ref. [3] shows that the stiffness matrix derived from the linear FVEM is a small perturbation of that of the linear CGFEM for a sufficiently small mesh size of the triangulation. An identity relating the stiffness matrix of linear FVEM and the

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matrix of linear CGFEM was established in [4]. A significant amount of work has been done to investigate the closeness of linear FVEM and linear CGFEM. However, the current understanding and implementation of higher order FVEMs is still at its infancy. For one-dimensional elliptic equations, high order FVEMs have been developed in [5]. Other relevant high order FVEM work can be found in [6–10].

As previously mentioned, the local variational formulation inherent in FVEM produces locally conservative fluxes, while, due to the global nature of its formulation, CGFEM does not. On the other hand, the robustness of CGFEM already has a firm mathematical foundation (see for example [11,12]) and its application to solve various problems is very extensive. Development of linear algebra solvers for CGFEM has reached an advanced stage, mainly driven from a solid understanding of the variational formulations and their properties, such as coercivity (and symmetry) of the bilinear form in the Galerkin formulation. This is not the case for the system derived from FVEMs, and in particular the linear algebraic system stemming from higher order FVEMs is ill-conditioned and difficult to solve. The matrix resulting from the FVEM is typically not symmetric. Furthermore, most discretization of FVEM using linear finite element basis yields an M-matrix. However, this is not true for FVEMs using the quadratic finite element basis (see [6]).

From an application point of view, the preservation of a local conservation property in the approximate solution is imperative for simulations of many physical problems, especially those that are derived from conservation laws. In order to maintain the advantages of CGFEM as well as to obtain locally conservative fluxes, post-processing techniques have been developed; see for example [13–31]. The post-processing techniques proposed in the aforementioned references are mainly designed for linear finite element methods and they include finite element methods for solving pure elliptic equations, advection diffusion equations, advection dominated diffusion equations, elasticity problems, Stokes problem, etc. Among them, some of the proposed post-processing techniques require solving auxiliary problems that are global in nature.

There currently exists a very limited amount of work aimed at developing post-processing techniques that produce locally conservative fluxes for high order CGFEMs. An interesting work on post-processing for high order CGFEMs is presented in [17]. This post-processing is performed by first solving a set of local systems, and then inverting a global system. A uniform approach to local reconstruction of local fluxes from various finite element method (FEM) solutions was presented in [32]. These methods include any order of conforming, nonconforming, and discontinuous FEMs. They proposed a hybrid formulation by utilizing Lagrange multipliers, which can be computed locally. However, the reconstructed fluxes are not locally conservative. Furthermore, they have used the reconstructed fluxes to derive an a posteriori error estimator; see [33].

The present investigation proposes a post-processing technique for CGFEM to obtain fluxes that are locally conservative across the boundaries of control volumes composing a dual mesh. In addition to its reliance on the original mesh, construction of this dual mesh is contingent on the polynomial degree of the CGFEM basis. The technique requires solving a low dimensional linear system for each finite element where local conservation is needed. Since each auxiliary problem can be solved independently, the proposed methodology can be readily implemented in modern parallel computing environments. Furthermore, due to this independence, the post-processing can be performed anywhere in the computational domain in which locally conservative fluxes are desired. While the forthcoming presentation utilizes a two-dimensional triangular mesh, the proposed technique is readily applicable to a rectangular mesh and three-dimensional meshes.

The rest of the paper is organized as follows. The CGFEM formulation of the model problem is presented in Section 2 followed by the description of the methodology of the post-processing technique in Section 3. Convergence analysis of the post-processing technique is presented in Section 4 and numerical examples are presented to demonstrate the performance of the proposed technique in Section 5.

2. Continuous Galerkin finite element method

As a model problem to be solved using the CGFEM we consider the elliptic boundary value problem

$$\begin{cases} -\nabla \cdot (\kappa \nabla u) = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases} \quad (2.1)$$

where Ω is a bounded open domain in $\mathbb{R}^2$ with Lipschitz boundary $\partial\Omega$, $\kappa = \kappa(\mathbf{x})$ is the elliptic coefficient, $u = u(\mathbf{x})$ is the solution to be found, $f = f(\mathbf{x})$ is a forcing function. Assuming $0 < \kappa_{\min} \leq \kappa(\mathbf{x}) \leq \kappa_{\max} < \infty$ for all $\mathbf{x} \in \Omega$ and $f \in L^2(\Omega)$, Lax–Milgram Theorem guarantees a unique weak solution to (2.1). On the assumption that Ω is polygonal, a partition $\mathcal{T}_h$ consisting of triangular elements τ such that $\overline{\Omega} = \bigcup_{\tau \in \mathcal{T}_h} \tau$ is constructed. In this setting, $h = \max_{\tau \in \mathcal{T}_h} h_\tau$ where h_τ is defined as the diameter of τ . The continuous Galerkin nodal finite element space is defined as

$$V_h^k = \{w_h \in C(\overline{\Omega}) : w_h|_\tau \in P^k(\tau), \forall \tau \in \mathcal{T}_h \text{ and } w_h|_{\partial\Omega} = 0\},$$

where $P^k(\tau)$ is a space of polynomials with degree at most k on τ . The CGFEM formulation for (2.1) is to find u_h with $(u_h - g_h) \in V_h^k$, such that

$$a(u_h, w_h) = \ell(w_h) \quad \forall w_h \in V_h^k, \quad (2.2)$$

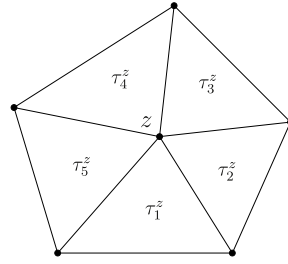


Fig. 1. Setting for V_h^1 : the dots represent the degrees of freedom and $\Omega^z = \cup_{i=1}^5 \tau_i^z$ is the support of ϕ_z .

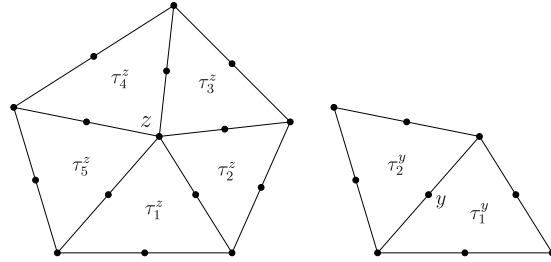


Fig. 2. Setting for V_h^2 : The dots represent the degrees of freedom. $\Omega^z = \cup_{i=1}^5 \tau_i^z$ (left) is support of ϕ_z and $\Omega^y = \cup_{i=1}^2 \tau_i^y$ (right) is the support of ϕ_y .

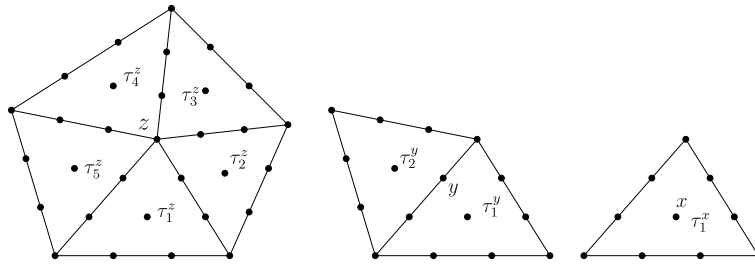


Fig. 3. Setting for V_h^3 : The dots represent the degrees of freedom. $\Omega^z = \cup_{i=1}^5 \tau_i^z$ (left) is the support of ϕ_z , and $\Omega^y = \cup_{i=1}^2 \tau_i^y$ (middle) is the support of ϕ_y , and $\Omega^x = \tau_1^x$ is the support of ϕ_x .

where

$$a(v, w) = \int_{\Omega} \kappa \nabla v \cdot \nabla w \, d\mathbf{x}, \quad \text{and} \quad \ell(w) = \int_{\Omega} f w \, d\mathbf{x},$$

and $g_h \in V_h^k$ can be thought of as the interpolant of g using the usual finite element basis. A drawback in this formulation, and motivation for the proposed methodology, is that the local conservation property is not imposed in the calculation of u_h . Consequently, a naive derivative calculation of u_h will not yield locally conservative fluxes. In fact, by construction of the finite element basis, ∇u_h will be discontinuous at the finite element's boundary. It is for this purpose that a post-processing of u_h must be performed.

The proposed post-processing crucially relies on the fact that u_h satisfies the CGFEM formulation. To make our discussion as general as possible, let Z be the set of nodes in Ω resulting from the partition $\mathcal{T}_h$ and placement of the degrees of freedom owned by V_h^k . In particular, Z consists of vertices for V_h^1 (see Fig. 1), vertices and degrees of freedom on the edges for V_h^2 (see Fig. 2), vertices and degrees of freedom on the edges and in the elements' barycenters for V_h^3 (see Fig. 3). Furthermore, $Z = Z_{\text{in}} \cup Z_{\text{d}}$, where Z_{in} is the set of interior degrees of freedom and Z_{d} is the set of corresponding nodes on $\partial\Omega$. Denoting the usual Lagrange nodal basis of V_h^k as $\{\phi_{\xi}\}_{\xi \in Z_{\text{in}}}$, (2.2) yields

$$a(u_h, \phi_{\xi}) = \ell(\phi_{\xi}) \quad \forall \xi \in Z_{\text{in}}. \quad (2.3)$$

For a $\xi \in Z_{\text{in}}$, let Ω^{ξ} be the support of the basis function ϕ_{ξ} . Then in the partition $\mathcal{T}_h$, $\Omega^{\xi} = \cup_{i=1}^{N_{\xi}} \tau_i^{\xi}$, where τ_i^{ξ} is an element that has ξ as one of its degrees of freedom, and N_{ξ} is the total number of such elements. In the linear case, N_{ξ} is the number of elements sharing the vertex ξ (see Fig. 1); in the quadratic case, N_{ξ} is either the number of elements sharing the vertex ξ or if ξ is a middle point on an edge then $N_{\xi} = 2$; (see Fig. 2); in the cubic case, N_{ξ} is determined

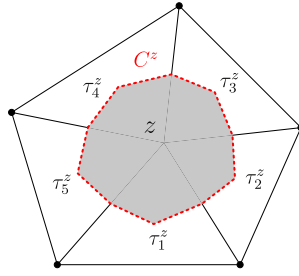


Fig. 4. C^z is the control volume corresponding to ϕ_z in V_h^1 .

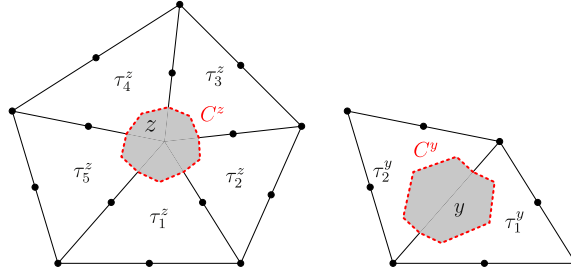


Fig. 5. C^z, C^y are control volumes corresponding to ϕ_z (left) and ϕ_y (right), respectively, in V_h^2 .

in the same way as in the quadratic case with the additional possibility that $N_\xi = 1$ if ξ is at the center of the element (see Fig. 3). With this in mind, (2.3) is expressed as

$$\sum_{i=1}^{N_\xi} a_{\tau_i^\xi}(u_h, \phi_\xi) = \sum_{i=1}^{N_\xi} \ell_{\tau_i^\xi}(\phi_\xi), \quad (2.4)$$

where $a_\tau(v_h, w_h)$ is $a(v_h, w_h)$ restricted to element τ and $\ell_\tau(w_h)$ is $\ell(w_h)$ restricted to element τ . Eq. (2.4) is fundamental and will be used in the derivation of the post-processing technique in Section 3.

3. A post-processing technique

This section lays out a post-processing technique to construct locally conservative fluxes over control volumes from CGFEM solutions. The focus is on a CGFEM solution u_h that is found in V_h^k with $k = 1, 2, 3$, i.e., linear, quadratic, and cubic CGFEMs, respectively. Construction for orders higher than these CGFEMs can be conducted in a similar fashion.

Based upon the original finite element mesh and V_h^k , a dual mesh that consists of control volumes is generated over which the post-processed fluxes are to be locally conservative. Let $\mathcal{D}_h$ be the collection of such control volumes. A control volume $C^\xi \in \mathcal{D}_h$ corresponds to a degree of freedom $\xi \in Z_{in}$. When a ξ is shared by more than one triangular elements, a C^ξ is constructed by connecting a set of polygons, each of which is a subregion of τ_i^ξ , $i = 1, \dots, N_\xi$. Otherwise, C^ξ is a proper subset of a triangular element.

The construction of C^ξ is contingent upon V_h^k . The process of creating the aforementioned polygons making up C^ξ is as follows. For V_h^1 , the triangular element's barycenter is connected to the middle points of the element's edges; see Fig. 4. For V_h^2 , the triangular element is first partitioned into four sub-triangles and then the barycenters and middle points of each sub-triangle are connected; see plots in Fig. 5. A similar procedure is used for V_h^3 , see plots in Fig. 6. A special case for V_h^3 is that there is one control volume that is a proper subset of a triangular element, which corresponds the ξ that is the barycenter of that element (see right most plot in Fig. 6). The whole process described in this paragraph is illustrated in Fig. 7.

Simply put, given a CGFEM solution $u_h \in V_h^k$, the post-processing aims at calculating

$$\tilde{u}_h = \sum_{\tau \in \mathcal{T}_h} \tilde{u}_{\tau,h} \chi_\tau \quad \text{such that} \quad \int_{\partial C^\xi} -\kappa \nabla \tilde{u}_h \cdot \mathbf{n} \, dl = \int_{C^\xi} f \, d\mathbf{x}, \quad \text{for every } C^\xi \in \mathcal{D}_h, \quad (3.1)$$

and $\kappa \nabla \tilde{u}_h \cdot \mathbf{n}$ is continuous on every boundary of C^ξ . Here χ_τ is the usual characteristic function of τ .

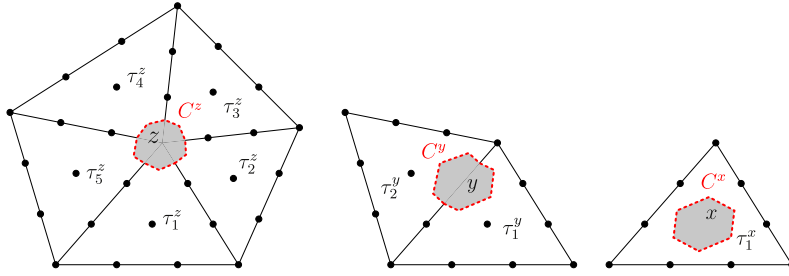


Fig. 6. C^z, C^y, C^x are control volumes corresponding to ϕ_z (left), ϕ_y (middle), and ϕ_x (right), respectively, in V_h^3 .

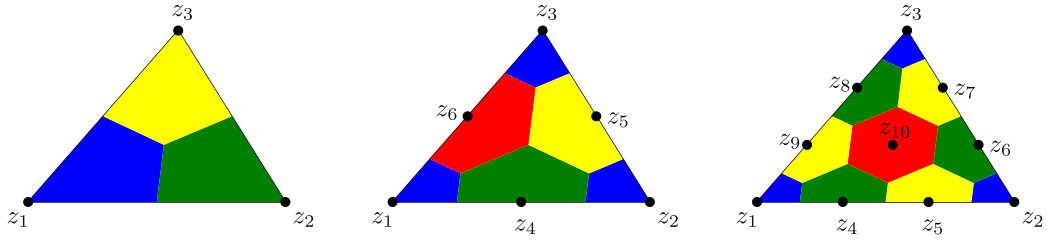


Fig. 7. Control volume construction and degrees of freedom on an element for V_h^k , $k = 1$ (left), $k = 2$ (middle), and $k = 3$ (right).

3.1. Auxiliary elemental problem

The procedure by which locally conservative fluxes on each control volume are produced relies on solving an elemental/local problem on τ . Let $N_k = \frac{1}{2}(k+1)(k+2)$ be the total number of degrees of freedom on a triangular element for V_h^k . Designate $s(\tau, k) = \{z_j\}_{j=1}^{N_k}$ the set of those degrees of freedom; see Fig. 7. As alluded to earlier, each triangular element τ is partitioned into N_k non-overlapping polygons, whose set is denoted by $\{t_{z_j}\}_{j=1}^{N_k}$; see Fig. 7. For t_ξ with $\xi \in s(\tau, k)$, the decomposition $\partial t_\xi = (\partial \tau \cap \partial t_\xi) \cup (\partial C^\xi \cap \partial t_\xi)$ is true.

Next, set

$$\{v\} = \frac{v|_{\tau_1} + v|_{\tau_2}}{2}. \quad (3.2)$$

to be the average of vector v on an edge or part of the edge that is shared by two elements τ_1 and τ_2 . Let $V^0(\tau)$ be the space of piecewise constant functions on element τ such that $V^0(\tau) = \text{span}\{\psi_\eta\}_{\eta \in s(\tau, k)}$, where ψ_η is the characteristic function of the polygonal t_η , i.e.,

$$\psi_\eta(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in t_\eta \\ 0 & \text{if } \mathbf{x} \notin t_\eta \end{cases}. \quad (3.3)$$

Define $I_\tau : H^1(\tau) \rightarrow V^0(\tau)$ to be a mapping such that $I_\tau w = \sum_{\xi \in s(\tau, k)} w_\xi \psi_\xi$, where $w_\xi = w(\xi)$ for $w \in H^1(\tau)$. Let $V_h^k(\tau) = \text{span}\{\phi_\eta\}_{\eta \in s(\tau, k)}$ where ϕ_η can be thought of as the usual nodal η basis function restricted to element τ . The elemental calculation for the post-processing is to find $\tilde{u}_{\tau, h} \in V_h^k(\tau)$ satisfying

$$b_\tau(\tilde{u}_{\tau, h}, w) = \ell_\tau(I_\tau w - w) + a_\tau(u_h, w) + e_\tau(u_h, I_\tau w - w), \quad \forall w \in V_h^k(\tau), \quad (3.4)$$

where

$$b_\tau(v, w) = \sum_{\eta \in s(\tau, k)} \int_{\partial C^\eta \cap \partial t_\eta} (-\kappa \nabla v \cdot \mathbf{n}) I_\tau w \, dl, \quad e_\tau(v, w) = \int_{\partial \tau} (\{\kappa \nabla v\} \cdot \mathbf{n}) w \, dl. \quad (3.5)$$

Lemma 3.1. The variational formulation (3.4) has a unique solution up to a constant.

Proof. By replacing the test function w with ϕ_ξ for $\xi \in s(\tau, k)$, (3.4) is reduced to

$$b_\tau(\tilde{u}_{\tau, h}, \phi_\xi) = \int_{t_\xi} f \, d\mathbf{x} - \ell_\tau(\phi_\xi) + a_\tau(u_h, \phi_\xi) + e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi), \quad \forall \xi \in s(\tau, k). \quad (3.6)$$

Recall that $\nabla \tilde{u}_{\tau,h}$ is continuous in τ and $\partial C^\eta \cap \partial t_\eta$ is a line inside τ . Thus, by assuming κ to be continuous in τ , it is true that

$$\sum_{\xi \in s(\tau,k)} b_\tau(\tilde{u}_{\tau,h}, \phi_\xi) = b_\tau(\tilde{u}_{\tau,h}, 1) = \sum_{\eta \in s(\tau,k)} \int_{\partial C^\eta \cap \partial t_\eta} -\kappa \nabla \tilde{u}_{\tau,h} \cdot \mathbf{n} \, dl = 0, \quad (3.7)$$

which implies that the left hand side of (3.6) is singular. Note that $\sum_{\xi \in s(\tau,k)} \phi_\xi = 1$ and the linearity of the $b_\tau(\cdot, \cdot)$ have been used in (3.6). By a similar argument,

$$\sum_{\xi \in s(\tau,k)} \ell_\tau(\phi_\xi) = \sum_{\xi \in s(\tau,k)} \int_\tau f \phi_\xi \, d\mathbf{x} = \int_\tau f \sum_{\xi \in s(\tau,k)} \phi_\xi \, d\mathbf{x} = \int_\tau f \, d\mathbf{x}.$$

Using the fact that $\nabla(\sum_{\xi \in s(\tau,k)} \phi_\xi) = \mathbf{0}$ and linearity of $a_\tau(\cdot, \cdot)$,

$$\sum_{\xi \in s(\tau,k)} a_\tau(u_h, \phi_\xi) = \sum_{\xi \in s(\tau,k)} \int_\tau \kappa \nabla u_h \cdot \nabla \phi_\xi \, d\mathbf{x} = \int_\tau \kappa \nabla u_h \cdot \nabla \left(\sum_{\xi \in s(\tau,k)} \phi_\xi \right) d\mathbf{x} = 0.$$

Furthermore,

$$\sum_{\xi \in s(\tau,k)} e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi) = \sum_{\xi \in s(\tau,k)} \int_{\partial \tau \cap \partial t_\xi} \{\kappa \nabla u_h\} \cdot \mathbf{n} \, dl - \int_{\partial \tau} \{\kappa \nabla u_h\} \cdot \mathbf{n} \sum_{\xi \in s(\tau,k)} \phi_\xi \, dl = 0.$$

Combining these equalities gives

$$\sum_{\xi \in s(\tau,k)} \left(\int_{\Gamma_\xi} f \, d\mathbf{x} - \ell_\tau(\phi_\xi) + a_\tau(u_h, \phi_\xi) + e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi) \right) = 0.$$

Thus despite the singularity of the left hand side of (3.6), the linear system in this equation is consistent. This consistency guarantees the existence of infinitely many solutions to (3.6), and thus also the solutions of (3.4). Moreover, for $\text{const} \in V_h^k$, it is straightforward to see that $b_\tau(\text{const}, w) = 0$. Now given $\tilde{u}_{\tau,h} \neq \tilde{v}_{\tau,h}$, each satisfying (3.4), then $\tilde{u}_{\tau,h} - \tilde{v}_{\tau,h}$ must satisfy $b_\tau(\tilde{u}_{\tau,h} - \tilde{v}_{\tau,h}, w) = 0$. But this means $\tilde{u}_{\tau,h} - \tilde{v}_{\tau,h} = \text{const}$. It is then concluded that the solution of (3.4) is unique up to a constant. $\square$

The preceding proof brings to light an interesting feature that is worth mentioning. If the only imperative is to construct a locally conservative flux in τ , one could have set the normal flux on $\partial \tau \cap \partial t_\xi$ as

$$\int_{\partial \tau \cap \partial t_\xi} \mathbf{v} \cdot \mathbf{n} \, dl = \ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi) - e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi).$$

Utilizing the arguments for the right hand side of (3.6) in the preceding proof,

$$\begin{aligned} \int_{\partial \tau} \mathbf{v} \cdot \mathbf{n} \, dl &= \sum_{\xi \in s(\tau,k)} \int_{\partial \tau \cap \partial t_\xi} \mathbf{v} \cdot \mathbf{n} \, dl \\ &= \sum_{\xi \in s(\tau,k)} \left(\ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi) - e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi) \right) \\ &= \int_\tau f \, d\mathbf{x}, \end{aligned}$$

which is indeed a statement of local conservation in τ . The main advantage of this simple approach is that it does not require solving any linear system. In the two-dimensional setting, $\partial \tau \cap \partial t_\xi$ consists of two segments like those shown in Fig. 7. Further refinement of the flux approximation on each segment, can be done, for example, by setting

$$\int_{\Gamma_\xi} \mathbf{v} \cdot \mathbf{n} \, dl = \frac{\ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi) - e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi)}{2} - \int_{\Gamma_\xi} \{\kappa \nabla u_h\} \cdot \mathbf{n} \, dl, \quad (3.8)$$

where Γ_ξ is a segment of $\partial \tau \cap \partial t_\xi$. If $\partial \tau \cap \partial t_\xi$ consists of more segments, the assignment of normal flux can be done, for instance, by putting relative weights to $\ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi) + e_\tau(u_h, \phi_\xi)$ according to the length ratio of the segment over $\partial \tau \cap \partial t_\xi$.

This simple approach unfortunately comes with a major drawback since the assigned normal flux is generally not continuous at a boundary of τ that is shared by another element τ' . One exception is when the values of $\ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi)$ from two neighboring elements are equal. This reveals the merits of the post-processing technique proposed in the present investigation. It truly provides a way to obtain locally conservative fluxes on control volumes that are continuous at the boundaries of each control volume.

Lemma 3.2. The true solution u of (2.1) satisfies

$$b_\tau(u, w) = \ell_\tau(I_\tau w - w) + a_\tau(u, w) + e_\tau(u, I_\tau w - w), \quad \forall w \in H^1(\tau), \quad (3.9)$$

and this further implies

$$b_\tau(u - \tilde{u}_{\tau,h}, w) = a_\tau(u - u_h, w) + e_\tau(u - u_h, I_\tau w - w), \quad \forall w \in V_h^k(\tau). \quad (3.10)$$

Proof. This is established by straightforward calculations. $\square$

The first result in Lemma 3.2 indicates that accurate boundary data makes it possible to obtain accurate fluxes. This is the very reason that the post-processing technique proposed in [15] and extended in [14,19,34] cannot be generalized to a post-processing technique for higher order CGFEMs. The boundary condition imposed in [15] is

$$\int_{\partial\tau \cap \partial t_\xi} -\kappa \nabla \tilde{u}_{\tau,h} \cdot \mathbf{n} \, dl = \ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi), \quad (3.11)$$

and clearly $\ell_\tau(w) - a_\tau(u, w) \neq 0$ for the true solution u of (2.1). Specifically for high order CGFEM solutions, only imposing $\ell_\tau(w) - a_\tau(u_h, w)$ as a boundary condition for $\tilde{u}_{\tau,h}$ is not sufficient to guarantee optimal accuracy. The convergence order and accuracy of the post-processed solution strongly depend on the boundary data. By imposing the boundary data $\ell_\tau(\phi_\xi) - a_\tau(u_h, \phi_\xi) - e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi)$, an optimal convergence order for the post-processed solution $\tilde{u}_h$ for any high order CGFEMs can be established.

Remark 1. Eq. (3.10) in Lemma 3.2 resembles Galerkin orthogonality and plays a crucial role in establishing the post-processing error.

3.2. Elemental linear system

Since the dimension of $V_h^k(\tau)$ is N_k , the variational formulation (3.4) yields an N_k -by- N_k linear algebraic system. Specifically, because $\tilde{u}_{\tau,h} \in V_h^k(\tau)$,

$$\tilde{u}_{\tau,h} = \sum_{\eta \in s(\tau, k)} \alpha_\eta \phi_\eta, \quad (3.12)$$

so insertion of this representation to (3.4) and substitution of the test function by ϕ_ξ give the linear algebraic system

$$A\alpha = \beta, \quad (3.13)$$

where $\alpha \in \mathbb{R}^{N_k}$ has entries that are the nodal solutions in (3.12), $\beta \in \mathbb{R}^{N_k}$ with entries

$$\beta_\xi = \ell_\tau(I_\tau \phi_\xi - \phi_\xi) + a_\tau(u_h, \phi_\xi) + e_\tau(u_h, I_\tau \phi_\xi - \phi_\xi), \quad \forall \xi \in s(\tau, k), \quad (3.14)$$

and A is a $N_k \times N_k$ matrix with entries

$$A_{\xi\eta} = b_\tau(\phi_\eta, \phi_\xi), \quad \forall \xi, \eta \in s(\tau, k), \quad (3.15)$$

As the linear system (3.13) is inherited from (3.4), it is singular but consistent, thus possessing a solution that is unique up to a constant (see Lemma 3.1). However, this does not cause any issue since to obtain locally conservative fluxes, the desired quantity from the post-processing is $\nabla \tilde{u}_{\tau,h}$, which is unique.

3.3. Local conservation

At this stage, the local conservation property (3.1) on control volumes for the post-processed solution is established. It is stated in the following lemma.

Lemma 3.3. The local conservation property (3.1) is satisfied in C^ξ where $\xi \in Z_{in}$.

Proof. Obviously, for $\xi = z_{10}$ in the case of V_h^3 , the polygonal $t_{z_{10}} = C^{z_{10}}$, (3.1) is directly satisfied from solving (3.4). A similar situation occurs for $k > 3$ in V_h^k . Thus, the statement of this lemma needs to be proven in the case that a control volume is associated with ξ that is either on the edge of τ or the vertex of τ . To reiterate previous notations, for a basis function ϕ_ξ , let $\Omega^\xi = \bigcup_{i=1}^{N_\xi} \tau_i^\xi$ be its support. Noting that the gradient component is averaged, it is obvious that

$$\sum_{j=1}^{N_\xi} \int_{\partial\tau_j^\xi} \{\kappa \nabla u_{\tau_j,h}\} \cdot \mathbf{n} \phi_\xi \, dl = 0 \quad \text{and} \quad \sum_{j=1}^{N_\xi} \int_{\partial\tau_j^\xi \cap \partial t_\xi} \{\kappa \nabla u_{\tau_j,h}\} \cdot \mathbf{n} \, dl = 0.$$

This implies that $\sum_{j=1}^{N_\xi} e_{\tau_j^\xi}(u_h, \phi_\xi) = 0$. A straightforward calculation and (2.4) give

$$\int_{\partial C^\xi} -\kappa \nabla \tilde{u}_{\tau,h} \cdot \mathbf{n} \, dl = \int_{C^\xi} f \, d\mathbf{x} + \sum_{j=1}^{N_\xi} \left(a_{\tau_j^\xi}(u_h, \phi_\xi) - \ell_{\tau_j^\xi}(\phi_\xi) - e_{\tau_j^\xi}(u_h, \phi_\xi) \right) = \int_{C^\xi} f \, d\mathbf{x},$$

which completes the proof. $\square$

4. An error analysis for the post-processing

In this section an error analysis of $\tilde{u}_h$ is performed that culminates in its optimal convergence property in H^1 semi-norm. As usual, $\|\cdot\|_{L^2}$ denotes the L^2 norm and $|\cdot|_W$ the usual semi-norm in a Sobolev space W . A property of the map I_τ defined in Section 3.1 is established as follows.

Lemma 4.1. *Let I_τ be as defined in Section 3.1. Then*

$$\|w - I_\tau w\|_{L^2(\tau)} \leq Ch_\tau^2 |w|_{H^2(\tau)} + Ch_\tau |w|_{H^1(\tau)}, \quad \text{for } w \in H^2(\tau). \quad (4.1)$$

Proof. For $w \in H^2(\tau)$, suppose $\Pi w \in V_h^1(\tau)$ is the standard linear interpolation of w so that $I_\tau w = I_\tau(\Pi w)$. By adding and subtracting Πw , and invoking triangle inequality,

$$\|w - I_\tau w\|_{L^2(\tau)} \leq \|w - \Pi w\|_{L^2(\tau)} + \|\Pi w - I_\tau(\Pi w)\|_{L^2(\tau)}.$$

Standard interpolation theory (see for example Theorem 4.2 in [35]) states that

$$\|w - \Pi w\|_{L^2(\tau)} \leq Ch_\tau^2 |w|_{H^2(\tau)}. \quad (4.2)$$

Since $I_\tau w = \sum_{\xi \in S(\tau,k)} w_\xi \psi_\xi$, τ can be partitioned equally into k^2 sub-triangles τ_j , $j = 1, \dots, k^2$. By Lemma 6.1 in [36],

$$\|\Pi w - I_\tau(\Pi w)\|_{L^2(\tau)}^2 = \sum_{j=1}^{k^2} \|\Pi w - I_\tau(\Pi w)\|_{L^2(\tau_j)}^2 \leq \sum_{j=1}^{k^2} Ch_{\tau_j}^2 |\Pi w|_{H^1(\tau_j)}^2 \leq Ch_\tau^2 |\Pi w|_{H^1(\tau)}^2.$$

Taking the square root gives

$$\|\Pi w - I_\tau(\Pi w)\|_{L^2(\tau)} \leq Ch_\tau |\Pi w|_{H^1(\tau)}. \quad (4.3)$$

By using triangle inequality and interpolation theory again,

$$|\Pi w|_{H^1(\tau)} \leq |w|_{H^1(\tau)} + |w - \Pi w|_{H^1(\tau)} \leq |w|_{H^1(\tau)} + Ch_\tau |w|_{H^2(\tau)}.$$

Combining these inequalities gives the desired result. $\square$

Lemma 4.2. *Fix a triangle $\tau = \tau_0$ and let $\{\tau_i\}_{i=1}^3$ be the neighbors (sharing edges) of τ , i.e., $\partial\tau \cap \partial\tau_i \neq \emptyset$. Then for $w, v \in H^2(\tau)$*

$$e_\tau(w, I_\tau v - v) \leq Ch_\tau^{1/2} \left(|v|_{H^1(\tau)} + h_\tau |v|_{H^2(\tau)} \right) \sum_{i=0}^3 \left(h_{\tau_i}^{-1/2} |w|_{H^1(\tau_i)} + h_{\tau_i}^{1/2} |w|_{H^2(\tau_i)} \right), \quad (4.4)$$

where C is a constant independent of h_τ and h_{τ_i} .

Proof. By definition and Cauchy–Schwarz inequality,

$$\begin{aligned} e_\tau(w, I_\tau v - v) &= \int_{\partial\tau} \{\kappa \nabla w\} \cdot \mathbf{n} (I_\tau v - v) \, dl \\ &\leq \left(\int_{\partial\tau} |\{\kappa \nabla w\} \cdot \mathbf{n}|^2 \, dl \right)^{1/2} \left(\int_{\partial\tau} |I_\tau v - v|^2 \, dl \right)^{1/2} \\ &\leq \kappa_{\tau, \max} \|\{\nabla w\} \cdot \mathbf{n}\|_{L^2(\partial\tau)} \|I_\tau v - v\|_{L^2(\partial\tau)}, \end{aligned} \quad (4.5)$$

where $\kappa_{\tau, \max}$ is the maximum of κ on τ . By trace inequality,

$$\begin{aligned} \|\{\nabla w\} \cdot \mathbf{n}\|_{L^2(\partial\tau)} &\leq \frac{1}{2} \sum_{i=0}^3 \|\nabla w \cdot \mathbf{n}\|_{L^2(\partial\tau_i)} \\ &\leq \frac{1}{2} \sum_{i=0}^3 \left(Ch_{\tau_i}^{-1/2} |w|_{H^1(\tau_i)} + Ch_{\tau_i}^{1/2} |w|_{H^2(\tau_i)} \right). \end{aligned} \quad (4.6)$$

Similarly by trace inequality and Lemma 4.1,

$$\begin{aligned}
 \|I_\tau v - v\|_{L^2(\partial\tau)} &\leq \sum_{\xi \in s(\tau, k)} \|I_\tau v - v\|_{L^2(\partial t_\xi)} \\
 &\leq \sum_{\xi \in s(\tau, k)} Ch_\tau^{-1/2} \|I_\tau v - v\|_{L^2(t_\xi)} + Ch_\tau^{1/2} |v|_{H^1(t_\xi)} \\
 &\leq C \left(h_\tau^{-1/2} \|I_\tau v - v\|_{L^2(\tau)} + h_\tau^{1/2} |v|_{H^1(\tau)} \right) \\
 &\leq Ch_\tau^{-1/2} \left(Ch_\tau^2 |v|_{H^2(\tau)} + Ch_\tau |v|_{H^1(\tau)} \right) + Ch_\tau^{1/2} |v|_{H^1(\tau)} \\
 &\leq Ch_\tau^{1/2} \left(|v|_{H^1(\tau)} + h_\tau |v|_{H^2(\tau)} \right),
 \end{aligned} \tag{4.7}$$

where t_ξ are the polygons defined in Fig. 7. Putting (4.6) and (4.7) into (4.5) then gives the desired result. $\square$

Lemma 4.3. Assume that $b_\tau(\cdot, \cdot)$ is bounded and coercive, i.e., for all $w \in H^2(\tau)$, $v \in V_h^k(\tau)$,

$$b_\tau(w, v) \leq C |w|_{H^1(\tau)} |v|_{H^1(\tau)} \text{ and } b_\tau(v, v) \geq C_b |v|_{H^1(\tau)}^2, \tag{4.8}$$

for some positive constants C and C_b . Then the following local error estimate holds true:

$$|u - \tilde{u}_{\tau, h}|_{H^1(\tau)} \leq C |u - u_h|_{H^1(\tau)} + Ch_\tau^{1/2} \sum_{i=0}^3 \left(h_{\tau_i}^{-1/2} |u - u_h|_{H^1(\tau_i)} + h_{\tau_i}^{1/2} |u - u_h|_{H^2(\tau_i)} \right).$$

Proof. Triangle inequality gives

$$|u - \tilde{u}_{\tau, h}|_{H^1(\tau)} \leq |u - u_h|_{H^1(\tau)} + |u_h - \tilde{u}_{\tau, h}|_{H^1(\tau)}. \tag{4.9}$$

Using the coercivity and boundedness of $b_\tau(\cdot, \cdot)$,

$$\begin{aligned}
 C_b |u_h - \tilde{u}_{\tau, h}|_{H^1(\tau)}^2 &\leq b_\tau(u_h - \tilde{u}_{\tau, h}, u_h - \tilde{u}_{\tau, h}) \\
 &= b_\tau(u_h - u, u_h - \tilde{u}_{\tau, h}) + b_\tau(u - \tilde{u}_{\tau, h}, u_h - \tilde{u}_{\tau, h}) \\
 &\leq C |u - u_h|_{H^1(\tau)} |u_h - \tilde{u}_{\tau, h}|_{H^1(\tau)} + b_\tau(u - \tilde{u}_{\tau, h}, u_h - \tilde{u}_{\tau, h}).
 \end{aligned} \tag{4.10}$$

For simplicity, let $\delta_{\tau, h} = u_h - \tilde{u}_{\tau, h}$. Then by (3.10),

$$b_\tau(u - \tilde{u}_{\tau, h}, \delta_{\tau, h}) = a_\tau(u - u_h, \delta_{\tau, h}) + e_\tau(u - u_h, \delta_{\tau, h}) \tag{4.11}$$

Now, by the boundedness of the bilinear form of $a_\tau(\cdot, \cdot)$,

$$a_\tau(u - u_h, \delta_{\tau, h}) \leq \kappa_{\tau, \max} |u - u_h|_{H^1(\tau)} |\delta_{\tau, h}|_{H^1(\tau)}, \tag{4.12}$$

where $\kappa_{\tau, \max}$ is the maximum of κ on τ . By Lemma 4.2 and inverse inequality,

$$\begin{aligned}
 e_\tau(u - u_h, \delta_{\tau, h}) &\leq Ch_\tau^{1/2} \left(|\delta_{\tau, h}|_{H^1(\tau)} + h_\tau |\delta_{\tau, h}|_{H^2(\tau)} \right) I \\
 &\leq Ch_\tau^{1/2} |\delta_{\tau, h}|_{H^1(\tau)} I.
 \end{aligned} \tag{4.13}$$

with

$$I = \sum_{i=0}^3 \left(h_{\tau_i}^{-1/2} |u - u_h|_{H^1(\tau_i)} + h_{\tau_i}^{1/2} |u - u_h|_{H^2(\tau_i)} \right).$$

Combining (4.10), (4.11), (4.12), and (4.13), dividing both sides by $|u_h - \tilde{u}_{\tau, h}|_{H^1(\tau)}$ and then putting it into (4.9) gives the desired result. $\square$

The boundedness of $b_\tau(\cdot, \cdot)$ has been established in Theorem 1 in [4]. Its coercivity is also established for V_h^1 (Theorem 2) and V_h^2 (Theorem 5) in [4]. To the best of the authors' knowledge there has been no formal analysis of the coercivity of the form for V_h^3 . In our numerical examples, however, we observe positive eigenvalues (hence coercivity) on the resulting matrices. We investigate the closeness of the FVEM bilinear form and the CGFEM bilinear form as follows.

The closeness of the linear FVEM and the linear CGFEM has been well studied. It has been shown in [3] that linear FVEM is a small perturbation of linear CGFEM. An identity between linear FVEM and linear CGFEM was established in [4]. However, for higher order FVEM and CGFEM, closeness has not been investigated as satisfactory as in the linear case. Now, as a by-product of our post-processing technique, we present and prove that for higher order case, the local FVEM bilinear form is close to its corresponding local CGFEM bilinear form with an error of order $\mathcal{O}(h_\tau)$.

Lemma 4.4. When κ is sufficiently smooth, we have

$$|b_\tau(v, w) - a_\tau(v, w)| \leq Ch_\tau(|v|_{H^1(\tau)} + |v|_{H^2(\tau)})|w|_{H^1(\tau)}, \quad (4.14)$$

for $v, w \in V_h^k(\tau)$.

Proof. Using definition (3.5) and the Divergence theorem, we have

$$b_\tau(v, w) = - \sum_{\xi \in v(\tau)} \int_{\partial C_\xi \cap \partial t_\xi} \kappa \nabla v \cdot \mathbf{n} I_\tau w \, dl = - \int_\tau \nabla \cdot (\kappa \nabla v) I_\tau w \, d\mathbf{x} + \int_{\partial\tau} \kappa \nabla v \cdot \mathbf{n} I_\tau w \, dl. \quad (4.15)$$

Furthermore, integration by parts gives

$$a_\tau(v, w) = - \int_\tau \nabla \cdot (\kappa \nabla v) w \, d\mathbf{x} + \int_{\partial\tau} \kappa \nabla v \cdot \mathbf{n} w \, dl. \quad (4.16)$$

Taking the difference of the two bilinear form yields

$$\begin{aligned} b_\tau(v, w) - a_\tau(v, w) &= \int_\tau \nabla \cdot (\kappa \nabla v)(w - I_\tau w) \, d\mathbf{x} + \int_{\partial\tau} \kappa \nabla v \cdot \mathbf{n}(I_\tau w - w) \, dl \\ &= I_1 + I_2 + I_3, \end{aligned} \quad (4.17)$$

where

$$\begin{aligned} I_1 &= \int_\tau \nabla \kappa \cdot \nabla v(w - I_\tau w) \, d\mathbf{x}, \\ I_2 &= \int_\tau \kappa \Delta v(w - I_\tau w) \, d\mathbf{x}, \\ I_3 &= \int_{\partial\tau} \kappa \nabla v \cdot \mathbf{n}(I_\tau w - w) \, dl. \end{aligned} \quad (4.18)$$

Using Cauchy–Schwarz inequality and applying Lemma 4.1, I_1 is estimated as

$$|I_1| \leq \|\nabla \kappa\|_{L^\infty(\tau)} |v|_{H^1(\tau)} \|w - I_\tau w\|_{L^2(\tau)} \leq C \|\nabla \kappa\|_{L^\infty(\tau)} h_\tau |v|_{H^1(\tau)} |w|_{H^1(\tau)}. \quad (4.19)$$

We also estimate I_2 in a similar fashion. Furthermore, we use the same argument in the proof of Lemma 4.2 and inverse inequality for estimation of I_3 , and use Lemma 4.1 to bound $\|w - I_\tau w\|_{L^2(\tau)}$. Combining these three estimates yields

$$|b_\tau(v, I_\tau w) - a_\tau(v, w)| \leq |I_1| + |I_2| + |I_3| \leq Ch_\tau(|v|_{H^1(\tau)} + |v|_{H^2(\tau)})|w|_{H^1(\tau)},$$

which completes the proof. $\square$

The local elemental problem (3.4) is a pure Neumann-type problem and Lemma 3.1 shows that its solution is unique up to a constant. The post-processed flux is unique. Thus, we establish the H^1 semi-norm error in the next theorem. An L^2 -norm error estimate of the post-processed solution is perhaps irrelevant due to the non-uniqueness of $\tilde{u}_h$. An appropriate condition must be taken that renders uniqueness of $\tilde{u}_h$. One condition is to choose a constant such that $\int_\tau \tilde{u}_h \, d\mathbf{x} = \int_\tau u_h \, d\mathbf{x}$. However, a rigorous analysis of this aspect is beyond the scope of the current investigation. For L^2 norm error estimate of a post-processed solution which uses local bubble functions, we refer to [37].

Theorem 4.1. Assume u in (2.1) is sufficiently smooth. Let $\tilde{u}_h = \sum_{\tau \in \mathcal{T}_h} \tilde{u}_{\tau,h} \chi_\tau$, where $\tilde{u}_{\tau,h} \in V_h^k(\tau)$ is the post-processed solution (3.12) and χ_τ is the usual characteristic function for τ . Then

$$|u - \tilde{u}_h|_{H^1(\Omega)} \leq Ch^k |u|_{H^{k+1}(\Omega)},$$

where C is a constant independent of h .

Proof. Noticing that $h_\tau \leq h$, this estimate can be proved by using Lemma 4.3 and arithmetic–geometric mean inequality:

$$\begin{aligned} |u - \tilde{u}_h|_{H^1(\Omega)}^2 &= \sum_\tau |u - \tilde{u}_h|_{H^1(\tau)}^2 \\ &\leq C \sum_\tau (|u - u_h|_{H^1(\tau)}^2 + h^2 |u - u_h|_{H^2(\tau)}^2) \\ &\leq C |u - u_h|_{H^1(\Omega)}^2 + Ch^2 |u - u_h|_{H^2(\Omega)}^2. \end{aligned} \quad (4.20)$$

It is well known (see Theorem 14.3.3 in [11] for example) that CGFEM approximations satisfy,

$$|u - u_h|_{H^1(\Omega)} \leq Ch^k |u|_{H^{k+1}(\Omega)}, \quad |u - u_h|_{H^2(\Omega)} \leq Ch^{k-1} |u|_{H^{k+1}(\Omega)}. \quad (4.21)$$

Substituting these inequalities back to (4.20) and taking the root gives the desired result. $\square$

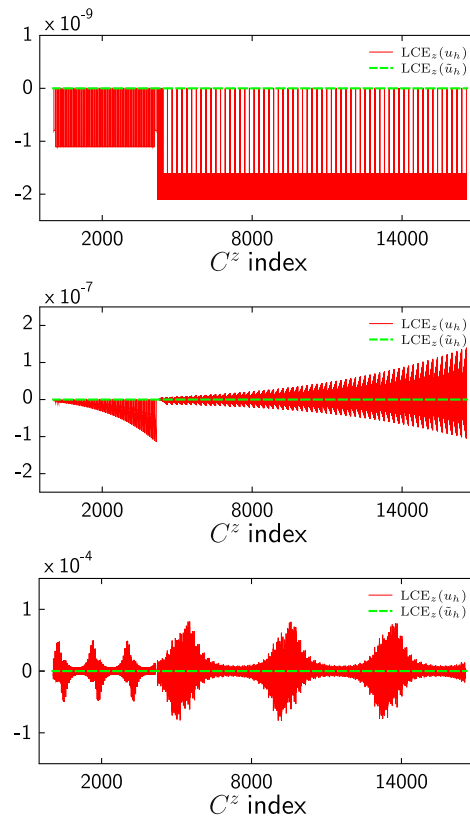


Fig. 8. $\text{LCE}_z(u_h)$ and $\text{LCE}_z(\tilde{u}_h)$ for Example 1 (top), Example 2 (middle), and Example 3 (bottom) using V_h^2 , $h = \sqrt{2}/64$.

5. Numerical experiments

Several numerical examples to illustrate the performance of the proposed post-processing technique are presented in this section. The main interest is to confirm the local conservation property of the post-processed fluxes and the convergence behavior of the post-processed solutions. With respect to the local conservation, the last experiment is on the simulation of flow and transport in a heterogeneous system.

5.1. Local conservation and convergence studies

The following test problems in the unit domain $[0, 1] \times [0, 1]$ are run for these studies.

- **Example 1.** Elliptic equation (2.1) with $\kappa = 1$, $u = (x - x^2)(y - y^2)$ with $g = 0$. The function f is derived from (2.1) that is satisfied by this known u .
- **Example 2.** Elliptic equation (2.1) with $\kappa = e^{2x-y^2}$, $f = -e^x$, $u = e^{-x+y^2}$ with g satisfying the true solution on $\partial\Omega$.
- **Example 3.** Elliptic equation (2.1) with

$$\kappa = \frac{1}{1 - 0.8 \sin(6\pi x)} \cdot \frac{1}{1 - 0.8 \sin(6\pi y)},$$

$$u = 1 - \frac{2 \cos(6\pi x) + 15\pi x - 2}{15\pi},$$

and $f = 0$. Dirichlet conditions $u(0, x_2) = 1$ (left boundary) and $u(1, x_2) = 0$ (right boundary) are imposed while the rest of $\partial\Omega$ (top and bottom boundaries) is set with a homogeneous Neumann condition.

The following metric is calculated to numerically illustrate the local conservation property of the post-processed fluxes as stated in (3.1):

$$\text{LCE}_z(\hat{u}_h) = \int_{\partial C^z} -\kappa \nabla \hat{u}_h \cdot \mathbf{n} \, dl - \int_{C^z} f \, d\mathbf{x}, \quad (5.1)$$

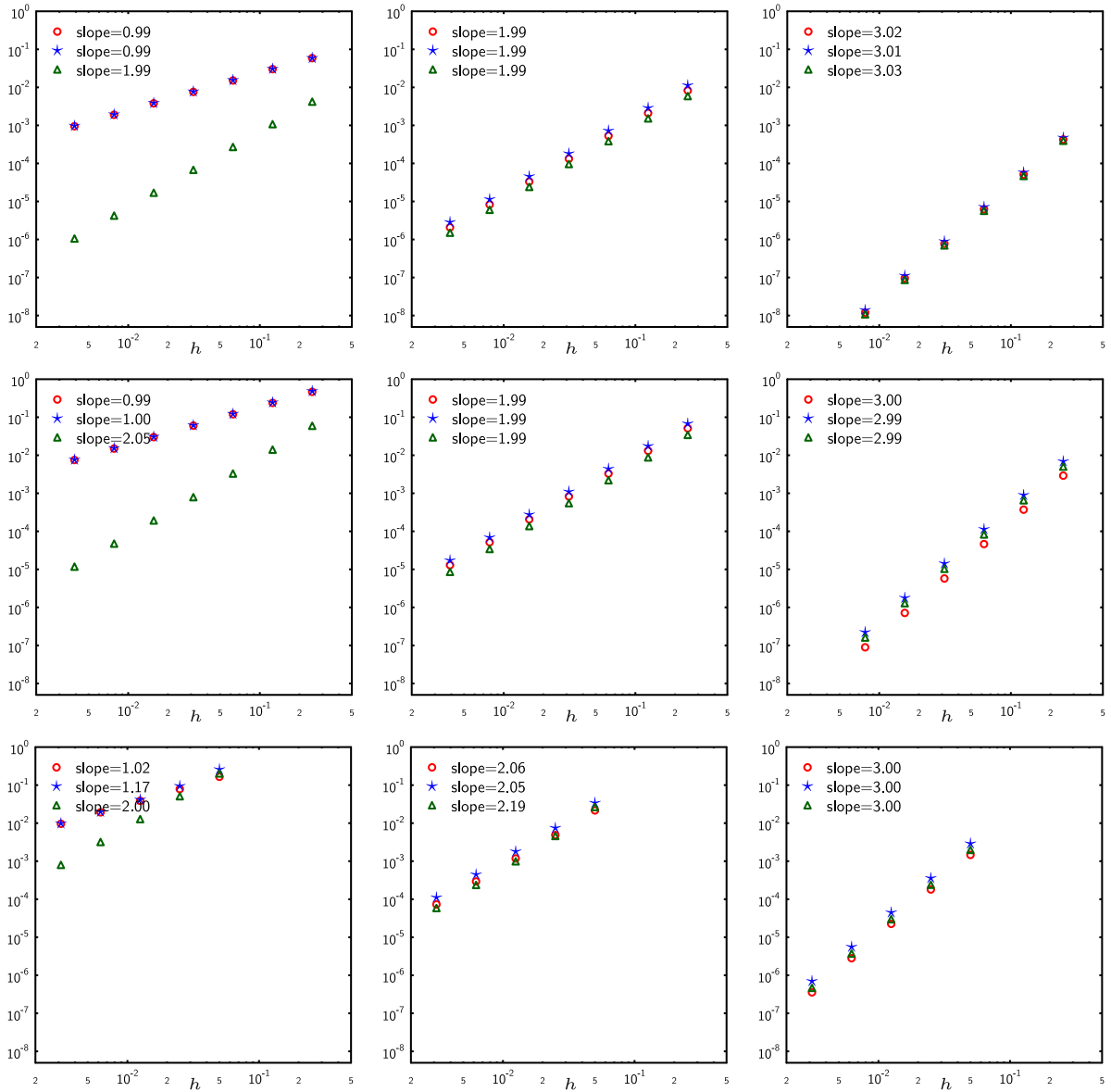


Fig. 9. H^1 semi-norm errors: $\circ$ is for $|u - u_h|_{H^1(\Omega)}$, $\star$ is for $|u - \tilde{u}_h|_{H^1(\Omega)}$, $\triangle$ is for $|u_h - \tilde{u}_h|_{H^1(\Omega)}$ for Example 1 (top row), 2 (middle row), and 3 (bottom row) using linear (left column), quadratic (middle column), and cubic (right column) CGFEM.

where $\hat{u}_h = u_h$ for CGFEM solution and $\hat{u}_h = \tilde{u}_h$ for the post-processed solution. The value of $\text{LCE}_z(\hat{u}_h)$ is plotted against the index of control volume C^z . Naturally, $\text{LCE}_z(\hat{u}_h) = 0$ means local conservation (3.1) is satisfied while $\text{LCE}_z(\hat{u}_h) \neq 0$ means local conservation (3.1) is not satisfied on the control volume C^z . Fig. 8 shows $\text{LCE}_z(\hat{u}_h)$ for the three examples above. Without a post-processing $\text{LCE}_z(u_h)$ are nonzero with u_h sought in V_h^2 . This means that the local conservation property (3.1) is not satisfied. The control volume indices in the figures are arranged as follows: indices from vertices of the mesh followed by the indices of the degrees of freedom on edges of elements. With the post-processing, $\text{LCE}_z(\tilde{u}_h)$ are shown to be practically negligible, which is mainly attributed to the errors in the application of numerical integration and the machine precision. Theoretically, these errors should be zeros as discussed in Section 3.3. As a side note, these plots give an indication that the magnitude of local conservation errors tends to get larger as the magnitude of κ increases. This is quite expected due to the proportionality of the flux to ∇u by this κ .

Next numerical convergence rates of the approximate solutions for Example 1, 2, and 3 are discussed. The H^1 semi-norm of $(u - u_h)$, $(u - \tilde{u}_h)$, and $(u_h - \tilde{u}_h)$ is computed. Fig. 9 depicts these values as they change with respect to h . The H^1 semi-norm errors of k th order CGFEM solutions and post-processed solutions are of optimal convergence orders, which confirm convergence analysis in Theorem 4.1 in Section 4.

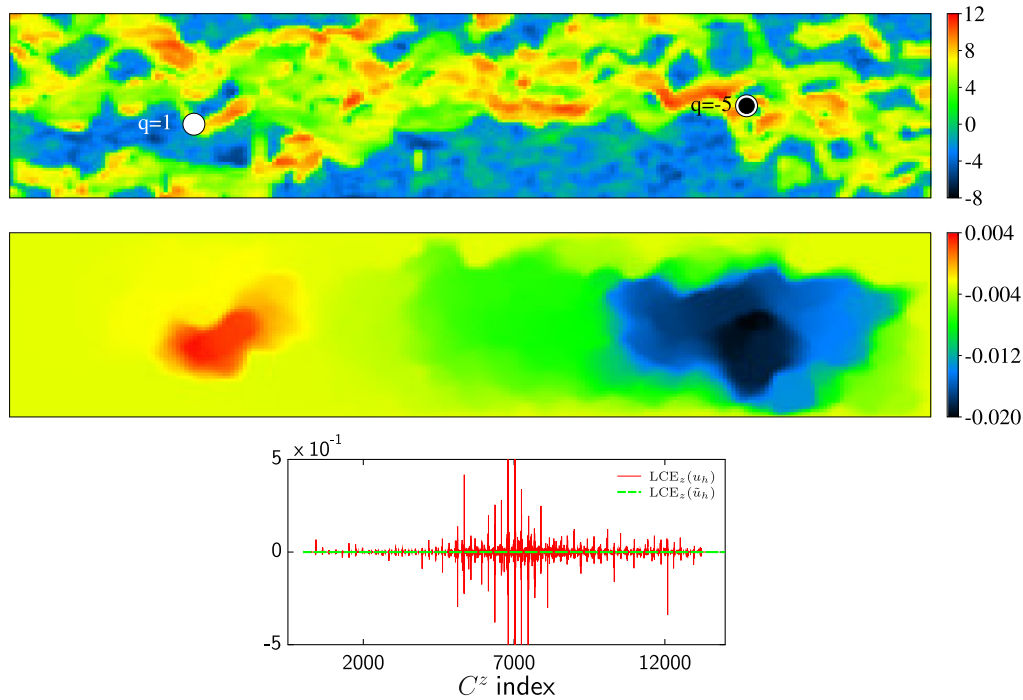


Fig. 10. Spatial profile of κ in logarithmic scale for Example 4 (top), $u_h \in V_h^1$ (middle), and $\text{LCE}_z(u_h)$ (bottom).

It is worth noting that the results presented in Fig. 9 suggest that $|u_h - \tilde{u}_h|_{H^1(\Omega)}$ is a reliable error estimator for high order CGFEM. In the linear case, $|u_h - \tilde{u}_h|_{H^1(\Omega)}$ is of order 2, which is higher than the optimal error convergence order of CGFEM solution.

5.2. Example 4: Simulation of flow and transport in a heterogeneous system

The numerical experiment in this subsection is intended to illustrate the cruciality of numerical local conservation in many simulations of problems that are derived from conservation laws. The choice is to investigate the problem of groundwater contamination by, say, pesticide, industrial solvent, or radioactive particles. The simplified model is to consider a natural groundwater flow through a spatially heterogeneous aquifer that is represented by the domain Ω . Contamination occurs when a certain unwanted chemical unexpectedly infiltrates the aquifer at a spill site and is transported with the groundwater to a leakage site. The contamination process is assumed to be modeled by a single-phase flow, i.e., the contaminated water is a single fluid whose essential character does not change with contaminant concentration.

Under the aforementioned assumption, Darcy's Law is applicable, which states that the fluid velocity is proportional to the fluid pressure gradient. Symbolically, the fluid velocity, $\mathbf{v}$, is given by $\mathbf{v}(\mathbf{x}) = -\kappa(\mathbf{x})\nabla u(\mathbf{x})$, where $u(\mathbf{x})$ denotes the pressure of the fluid. The coefficient of proportionality $\kappa(\mathbf{x})$ in the above relation denotes the permeability of the aquifer, which exhibits spatially disparate magnitudes.

Moreover, let $S(\mathbf{x}, t)$ denote the dimensionless contaminant concentration whose values are confined between 0 and 1. It is assumed that mass fluxes due to dispersion and diffusion are small relative to advective mass flux, and the fluid–solid interface does not permit the fluid to cross it. Taking the balance of the Darcy's velocity against source terms and combining it with the dynamics of $S(\mathbf{x}, t)$ governed by the hyperbolic partial differential equation yields (see for example [38])

$$\begin{cases} -\nabla \cdot (\kappa \nabla u) = q & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \\ \partial_t S - \nabla \cdot (f(S)\kappa \nabla u) = qS & \text{in } \Omega \times [0, T], \\ S(\mathbf{x}, 0) = 0 & \text{on } \Omega. \end{cases} \quad (5.2)$$

The model in (5.2) is a typical one way coupled system in which u is first solved, and in turn is used as an input to the concentration equation to get S . Related two-way coupling model to illustrate an application of the post-processing has previously been presented in [39]. It is assumed that u is prescribed with a homogeneous Dirichlet condition and there is no contaminant in the aquifer before the spill.

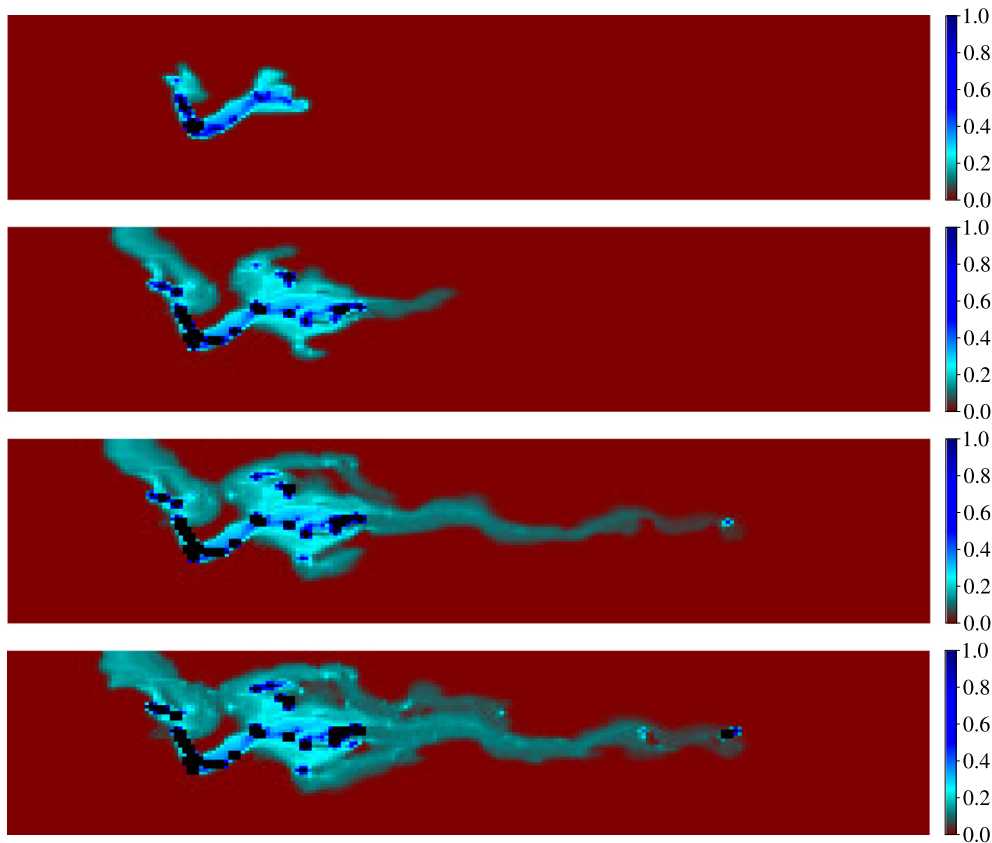


Fig. 11. Spatial profile of $S_h^n \in V_h^0$ at $t_n = 0.1, 0.5, 1, 2$ (top-bottom); $u_h \in V_h^1$ without post-processing is used.

To conduct the simulation, the aquifer is chosen to be $\Omega = (0, 5) \times (0, 1)$, whose capability to flow fluid is quantified by $\kappa(\mathbf{x})$ shown in the top plot of Fig. 10. As indicated in this plot, the magnitude of κ varies sharply over a variety of length scales. In particular, it forms a channelized feature in which a subregion of Ω has a much larger value of κ compared its surroundings. This feature gives a distinct effect in the transport dynamics of S in that the channelized subregion is the preferential flow regime. Furthermore, source term is $q(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}_{\text{in}}) - 5\delta(\mathbf{x} - \mathbf{x}_{\text{out}})$, which represents a rate of spill at $\mathbf{x}_{\text{in}} = (1, 0.4)$ and a rate of leakage at $\mathbf{x}_{\text{out}} = (4, 0.5)$ (see top plot of Fig. 10). A linear CGFEM with a 220×60 rectangular mesh is used to solve for u_h , whose spatial profile is shown in the middle plot of Fig. 10.

The contaminant infiltrating the aquifer at the spill site has a concentration $S(\mathbf{x}_{\text{in}}, t) = 1$. The concentration equation in (5.2) is approximated by a time-explicit nodal-centered finite volume method:

$$\int_{C^z} \frac{S_h^{n+1} - S_h^n}{\Delta t} d\mathbf{x} = \int_{\partial C^z} f(S_h^n) \kappa \nabla \hat{u}_h \cdot \mathbf{n} d\ell + \int_{C^z} q S_h^n d\mathbf{x} \quad \forall C^z \in \mathcal{D}_h, \quad (5.3)$$

where $S_h^n \approx S(\cdot, t_n)$ with $t_n = n\Delta t$, $n = 1, 2, \dots$. A time step $\Delta t = 10^{-4}$ is used in the simulation. In the spatial dimension it is assumed that $S_h^n = \sum_{z \in \mathcal{Z}} S_z^n \chi_{C^z}$ for real numbers S_z^n , i.e., a piecewise constant function over $\mathcal{D}_h$. An upwinding scheme with a minmod limiter (see for example [40]) is applied in the first term of the right hand side of (5.3).

At this stage there are two ways to compute $\kappa \nabla \hat{u}_h \cdot \mathbf{n}$: using a naive calculation of $\kappa \nabla u_h \cdot \mathbf{n}$ or applying the post-processing to get $\kappa \nabla \tilde{u}_h \cdot \mathbf{n}$ that has been designed to have the local conservation property. As rendered in the bottom plot of Fig. 10, the corresponding local conservation error of $\kappa \nabla u_h \cdot \mathbf{n}$ is quite large. Recall that the expression of the first term of the right hand side of (5.3) resembles a statement of local conservation for the Darcy's velocity. In recognition of this expression, the latter alternative is the more appropriate approach. Fig. 11 shows the spatial profiles of S_h at several time levels obtained from using a naive calculation of $\kappa \nabla u_h \cdot \mathbf{n}$ as an input. Indeed using u_h without a post-processing yields unphysical concentration profiles, in particular there are regions on which the concentration is above 1 indicated by black spots in the plots. This unphysical behavior is attributed to the lack of local conservation of u_h and spreads out throughout the aquifer, yielding an overall unreliable prediction of the numerical model. Fig. 12 shows the concentration profiles when the post-processing technique to get $\kappa \nabla \tilde{u}_h \cdot \mathbf{n}$ is used as an input. In contrast to the previous result, the profile maintains a value between 0 and 1 for the whole time span of simulation. This example undoubtedly demonstrates the necessity of having the local conservation property be satisfied by the approximate solution.

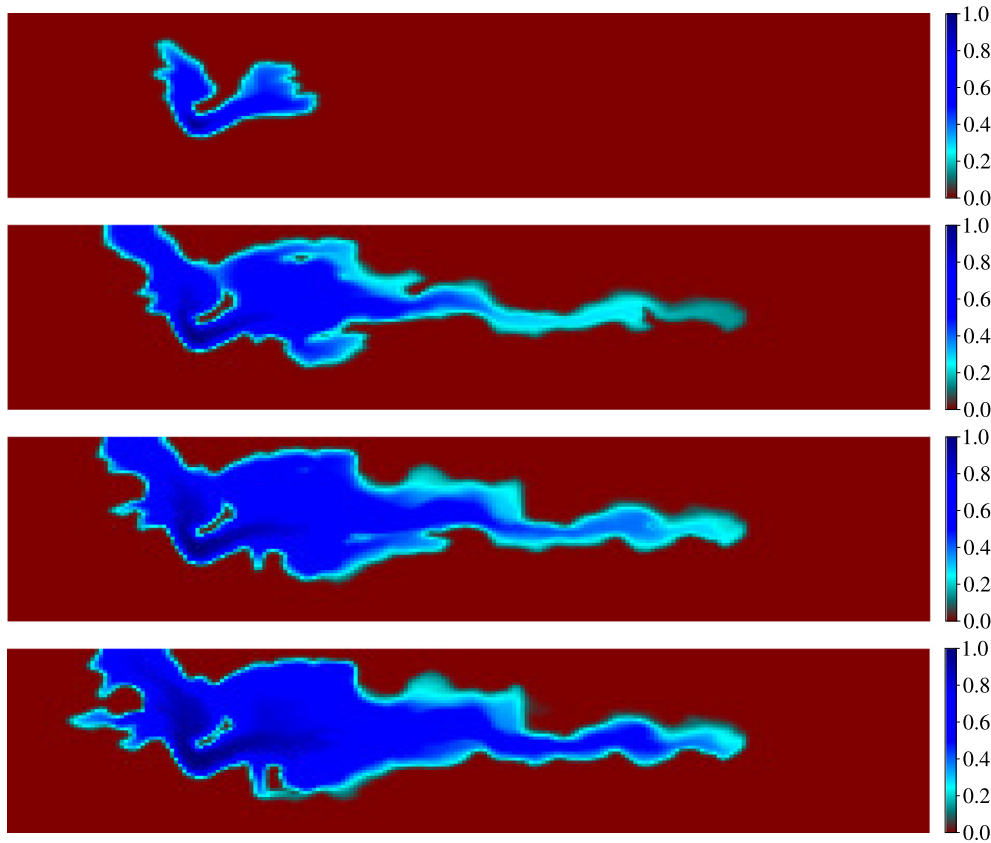


Fig. 12. Spatial profile of $S_h^n \in V_h^0$ at $t_n = 0.1, 0.5, 1, 2$ (top–bottom); $\tilde{u}_h \in V_h^1$ (the post-processing) is used.

6. Conclusion

The present investigation proposes a post-processing technique for any order CGFEM used to solve elliptic boundary value problems. This technique builds a bridge between any order FVEM and any order of CGFEM. Although FVEMs have the advantage of producing approximate solutions that satisfy the local conservation property, they still lack theoretical foundations (especially for higher order FVEMs), and solving the resulting algebraic systems presents its own challenges. On the other hand CGFEMs already stand on a sure-footed theory and the resulting algebraic system is in general more amenable to solve. However, its approximate solutions found with CGFEMs lack the local conservation property. In this less than ideal situation the proposed post-processing technique can serve as a great tool if one would like to use the CGFEM to solve PDEs where the local conservation property is a necessity.

Since elemental problems essential in the post-processing technique are localized, they are independent of each other. For linear CGFEM, the technique requires solving a 3-by-3 system for each element while for quadratic and cubic CGFEMs, the technique requires solving a 6-by-6 system and 10-by-10 system for each finite element, respectively. It thus can be easily, naturally, and efficiently implemented in a parallel computing environment.

As for future work, one can use this technique for other differential equations, such as advection diffusion equations, two-phase flow problems, and elasticity models; also one can apply this technique for other numerical methods, such as SUPG. One interesting direction is to develop a post-processing technique which requires solving only 3-by-3 systems for each element and for any order of CGFEMs.

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